

PERMANENT TEETH | CLINICAL QUICK REFERENCE

Vital Pulp Therapy

VPT succeeds when the right tooth is selected, the pulp is reassessed after exposure, diseased tissue is removed to a controllable level, asepsis is maintained, and the tooth is sealed and reviewed.

SELECT THE TOOTH

Patient and pain

Record provoked, spontaneous, nocturnal, and lingering pain, previous episodes, analgesic use, medical factors, expectations, and ability to attend review.

Pulp and apical tests

Compare cold and EPT with control teeth. Add percussion, palpation, mobility, periodontal probing, and crack assessment.

Imaging and caries depth

Assess the dentine barrier, root maturity, resorption, and apical status on a diagnostic periapical radiograph.

Restorability and isolation

Confirm a defensible periodontal and restorative prognosis, reliable rubber-dam isolation, sound margins, and a durable final restoration.

Consent and conversion

Discuss indirect pulp treatment, direct capping, partial or full pulpotomy, root-canal treatment, extraction, and possible intra-operative conversion.

CHOOSE THE ROUTE

1. Deep caries with a dentine barrier

For normal pulp or reversible pulpitis, selective caries removal and indirect pulp treatment can preserve the barrier while sound peripheral dentine supports the seal.

2. Carious pulp exposure

Direct pulp capping may suit a carefully selected small exposure. Partial pulpotomy removes superficial inflamed tissue and can provide a cleaner endpoint.

3. Extremely deep or symptomatic caries

Complete excavation permits direct assessment. If the pulp is vital, the apical tissues are suitable, and bleeding becomes controllable, full pulpotomy may preserve the radicular pulp.

4. Traumatic exposure

Exposure size, contamination, delay, root maturity, and associated injuries guide care. Current guidance favours partial or full pulpotomy over direct capping in many traumatic exposures.

STOP VPT when necrosis, purulence, infection-related apical involvement, or uncontrolled bleeding after appropriate pulpotomy prevents a defensible tissue endpoint. Reassess for pulpectomy, root-canal treatment, or referral.

OPERATIVE SEQUENCE

Keep the caries phase separate from the pulp phase

The exposed pulp is a surgical wound. The tissue remains part of the diagnosis throughout the procedure.

01 Isolate and disinfect

Achieve profound anaesthesia. Place rubber dam before caries removal and disinfect the crown, clamp, and adjacent dam.

02 Create sound margins

Remove caries according to the selected route. Establish clean peripheral dentine that can support a reliable adhesive seal.

03 Change to the pulp set

Before entering the wound, change to sterile burs, instruments, and sterile coolant. Do not carry a caries-contaminated bur into the pulp.

04 Inspect the tissue

Under magnification, look for uniformly vital tissue, focal necrosis, purulence, clot, debris, and the character of bleeding.

05 Remove tissue with purpose

Resect to a level where the remaining pulp appears healthy and the wound can be controlled. Let the tissue determine depth.

06 Test the endpoint

Use sodium hypochlorite according to the clinical protocol. A clean, controllable wound can receive cement; persistent uncontrolled bleeding changes the treatment.

HAEMOSTASIS IS AN ASSESSMENT

Inspect

Direct observation is the second diagnostic stage. Confirm vitality and identify necrosis, purulence, clot, or debris.

Control

Apply sodium hypochlorite by passive irrigation or a lightly moistened sterile pellet, with gentle pressure and appropriate soft-tissue protection.

Reassess

At two-minute checkpoints, remove the pellet and inspect. Extend a partial pulpotomy only when the tissue condition requires further removal.

Decide by six minutes

Current AAPD guidance associates haemostasis within six minutes with higher pulpotomy success. Uncontrolled bleeding after full pulpotomy ends the VPT pathway.

A full pulpotomy removes the chamber pulp to the canal orifices. It often provides a clearer field for assessing the radicular pulp. Place calcium-silicate cement only when the wound is clean and controlled.

SEAL AND REVIEW

Protect the tissue that remains

Material choice follows selection, asepsis, tissue assessment, and haemostasis. The coronal restoration is part of the pulp treatment.

07 Calcium-silicate cement

Place a validated hydraulic calcium-silicate cement over the controlled wound. Follow the IFU for thickness, moisture, setting, overlay, and staining risk.

08 Definitive restoration

Complete a well-sealed restoration at the same visit where feasible. Use compatible materials and plan cuspal protection from the remaining structure and fracture risk.

09 Record

Document the diagnosis, exposure, pulpotomy level, tissue findings, haemostasis time, material and lot, restoration, consent, and baseline image.

10 Review

Review clinically at six months, clinically and radiographically at 12 months, then according to the case and current guidance. Reassess sooner if symptoms develop.

SUCCESS AND FAILURE

Findings consistent with success

Comfortable, functional tooth. No swelling, sinus tract, pathological tenderness, new apical disease, or internal resorption. Continued root development when immature.

Prompt reassessment

Persistent or worsening spontaneous pain, swelling, sinus tract, tenderness, loss of function, new apical change, internal resorption, or failure of the coronal seal.

Cold and EPT may help after direct capping or partial pulpotomy. A full pulpotomy may not respond even when radicular pulp remains vital; judge the case from symptoms, examination, and radiographs.

CHAIRSIDE RECORD

☐ Diagnosis, restorability, and alternatives recorded	☐ Rubber dam and magnification used
☐ Sterile pulp-phase burs and coolant used	☐ Tissue findings and haemostasis time recorded
☐ Calcium-silicate product and lot recorded	☐ Definitive seal completed or documented
☐ Baseline clinical/radiographic record retained	☐ Review interval and failure advice given

SOURCES

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4. Benjamin G, Gulati J. Pulpotomies for Irreversible Pulpitis? The Rise of Vital Pulp Therapy. *Protrusive Dental Podcast, PDP133*. 2022.
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